

1. **MS:** 2.1-2.9, 2.15-2.29, 2.31, 3.2-3.3, 3.5-3.9

2. From Sydsaeter and Hammond

1.7.3. Determine x such that:

(a) $|3 - 2x| = 5$

(b) $|x| \leq 2$

(c) $|x - 2| \leq 1$

(d) $|3 - 8x| \leq 5$

(e) $|x| > \sqrt{2}$

(f) $|x^2 - 2| \leq 1$

2.4.1 Solve the systems of equations:

(a) $x - y = 5$
 $x + y = 11$

(b) $4x - 3y = 1$
 $2x + 9y = 4$

(c) $3x + 4y = 2.1$
 $5x - 6y = 7.3$

4.2.1. Let $f(x) = x^2 + 1$

(a) Compute $f(0)$, $f(\frac{1}{2})$, and $f(\sqrt{2})$.

(b) For what values of x is it true that

(i) $f(x) = f(-x)$

(ii) $f(x + 1) = f(x) + f(1)$

(iii) $f(2x) = 2f(x)$

4.3.3.

x	0	1	2	3	4
$g(x) = -2x + 5$					

4.3.4

x	-2	-1	0	1	2	3	4
$h(x) = x^2 - 2x - 3$							

4.10.2 Solve the following equations for x :

(a) $3^x = 8$

(d) $\ln[x(x-2)] = 0$

(b) $\ln x = 3$

(e) $\frac{x(x+3)}{x^2+1} = 0$

(c) $\ln(x^2 - 4x + 5) = 0$

(f) $\ln(\sqrt{x} - 5) = 0$

3. Additional Questions:

- (a) An environmental study of a certain community suggests that the average daily level of carbon monoxide in the air will be $c(p) = 0.5p + 1$ parts per million when the population is p thousand. It is estimated that t years from now the population of the community will be $p(t) = 10 + 0.1t^2$.
- (i) Express the level of carbon monoxide in the air as a function of time, i.e., find $c(p(t))$.
 - (ii) When will the carbon monoxide level reach 6.8 parts per million?
- (b) You have been asked to analyze the steel market. From public sources, you are able to find that last year's price for steel was \$20 per ton. At this price, 100 million tons were sold on the world market. From trade association data, you can obtain estimates for the own-price elasticities of demand and supply on the world markets as -0.25 for demand and 0.5 for supply, respectively. Assume that steel has linear demand and supply curves throughout. Find the demand and supply equations in this market and sketch the demand and supply curves.